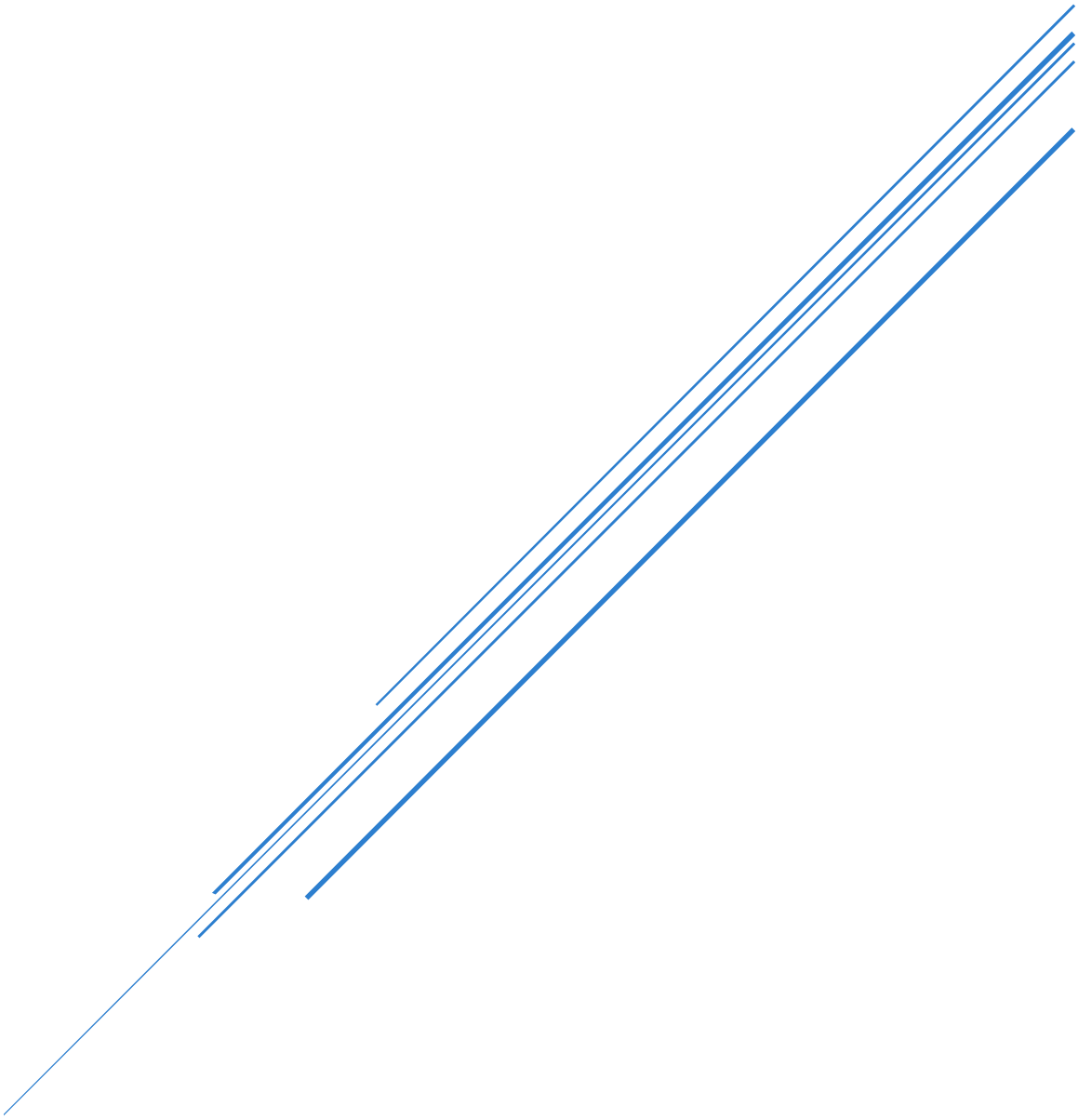
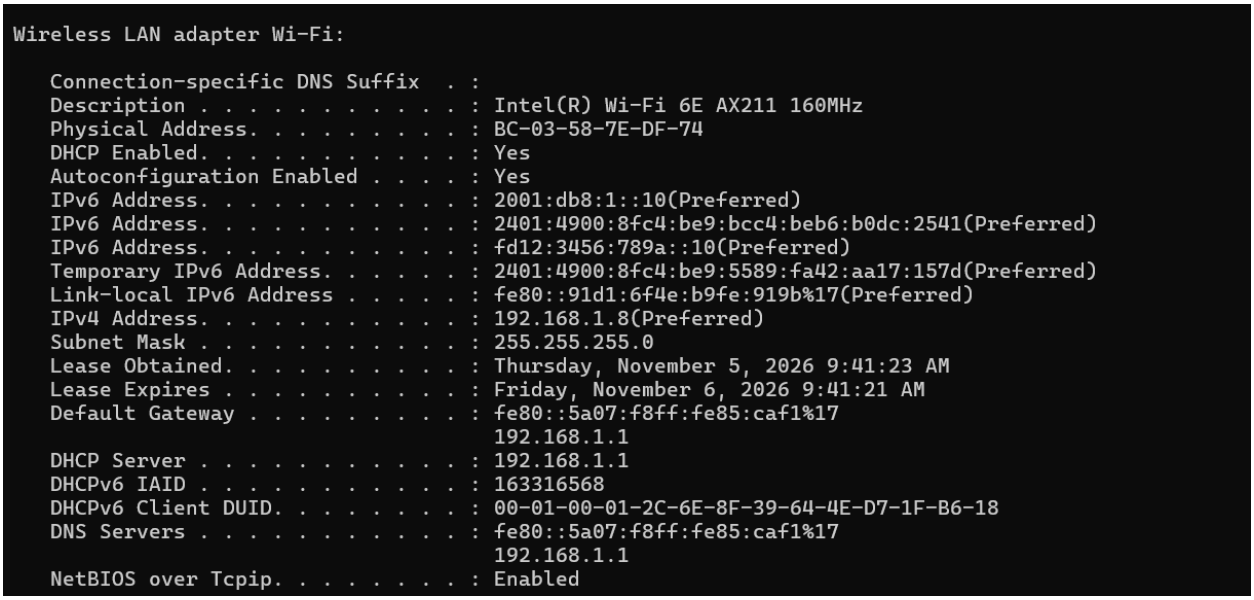
NETWORK TROUBLESHOOTING, MANAGEMENT, MONITORING, AND OPTIMIZATION

Assignment - 8



1. **Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.**

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```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
   Physical Address. . . . . . . . . : BC-03-58-7E-DF-74
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:db8:1::10(Preferred)
   IPv6 Address. . . . . . . . . . . : 2401:4900:8fc4:be9:bcc4:beb6:b0dc:2541(Preferred)
   IPv6 Address. . . . . . . . . . . : fd12:3456:789a::10(Preferred)
   Temporary IPv6 Address. . . . . . : 2401:4900:8fc4:be9:5589:fa42:aa17:157d(Preferred)
   Link-local IPv6 Address . . . . . : fe80::91d1:6f4e:b9fe:919b%17(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.8(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, November 5, 2026 9:41:23 AM
   Lease Expires . . . . . . . . . . : Friday, November 6, 2026 9:41:21 AM
   Default Gateway . . . . . . . . . : fe80::5a07:f8ff:fe85:caf1%17
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 163316568
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2C-6E-8F-39-64-4E-D7-1F-B6-18
   DNS Servers . . . . . . . . . . . : fe80::5a07:f8ff:fe85:caf1%17
                                       192.168.1.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

After executing the command in Windows Command Prompt, the following network details were obtained from the Wi-Fi adapter.

Network Configuration Details

Parameter	Result
IPv4 Address	192.168.1.8
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.1

2. **Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.**

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```
C:\Users\RAHUL>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Users\RAHUL>ping www.google.com

Pinging www.google.com [2001:4860:482c:7700::] with 32 bytes of data:
Reply from 2001:4860:482c:7700::: time=57ms
Reply from 2001:4860:482c:7700::: time=77ms
Reply from 2001:4860:482c:7700::: time=53ms
Reply from 2001:4860:482c:7700::: time=57ms

Ping statistics for 2001:4860:482c:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 53ms, Maximum = 77ms, Average = 61ms
```

1. When Wi-Fi Was Disconnected

Output:

Ping request could not find host www.google.com.

Please check the name and try again.

2. When Wi-Fi Was Reconnected

Output:

Reply from 2001:4860:482c:7700:::time=57ms

Reply from 2001:4860:482c:7700:::time=77ms

Reply from 2001:4860:482c:7700:::time=53ms

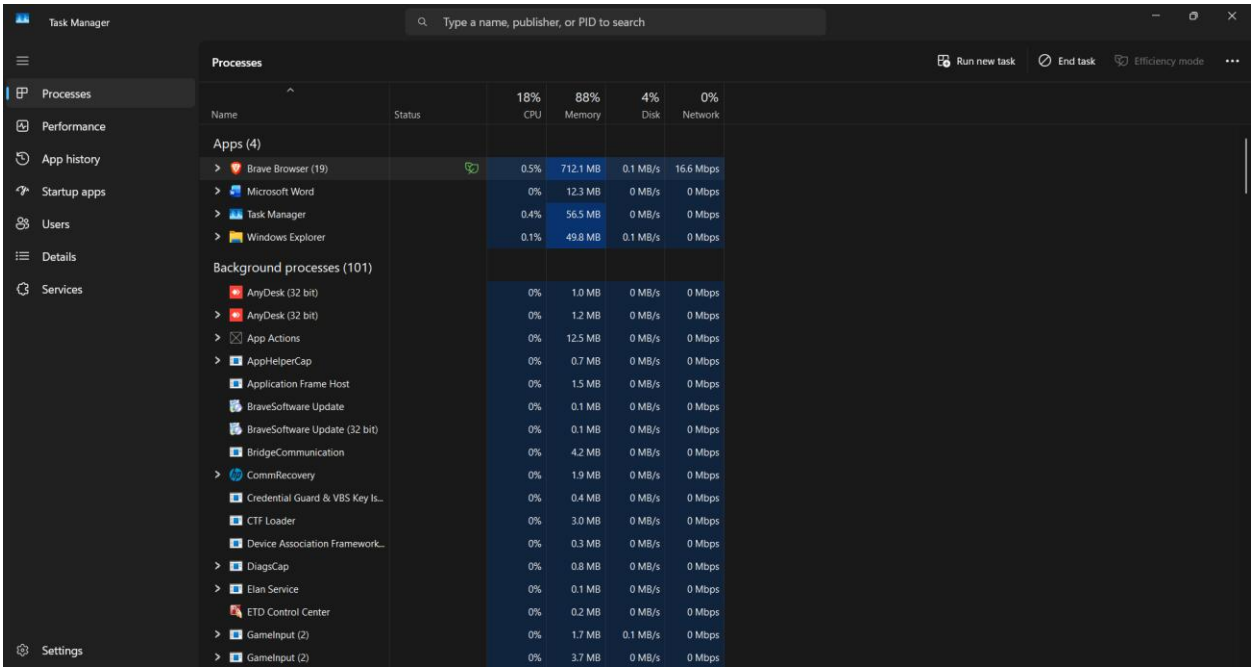
Reply from 2001:4860:482c:7700:::time=57ms

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Minimum = 53ms, Maximum = 77ms, Average = 61ms

3. **Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.**

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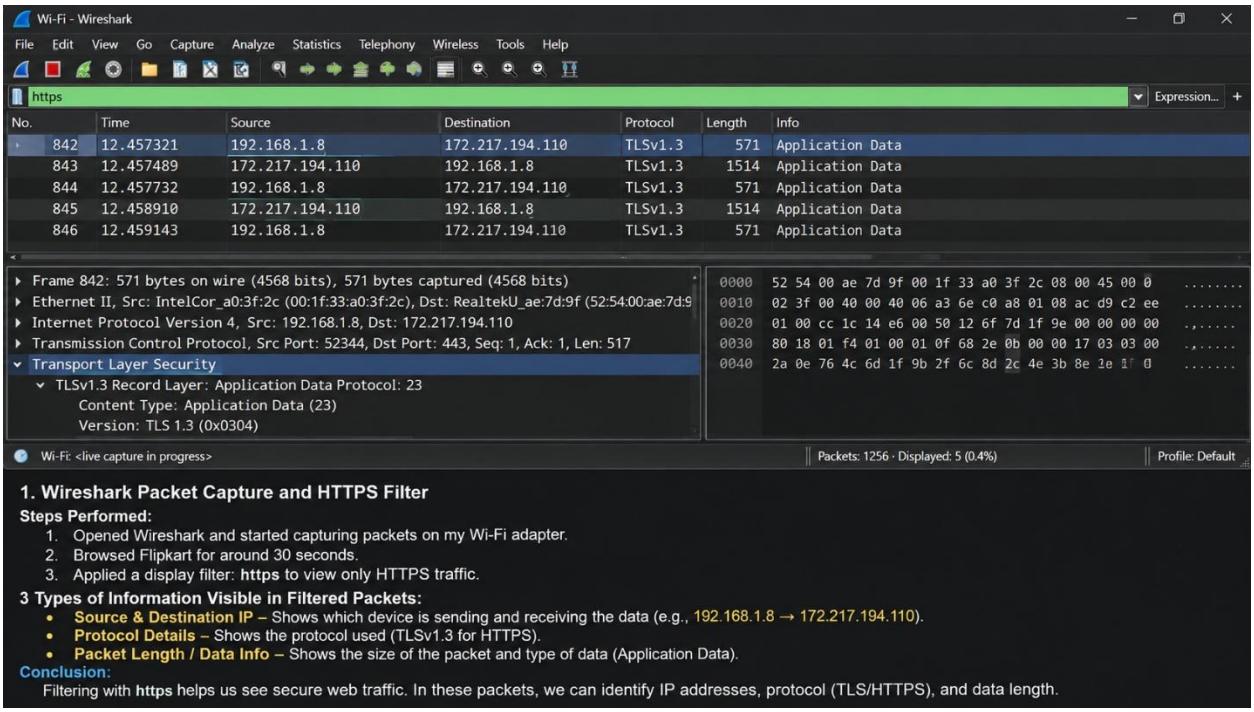


During YouTube video streaming, the Brave Browser consumed the highest network bandwidth because it was continuously downloading video and audio data from YouTube servers.

The screenshot shows Brave Browser using 16.6 Mbps of network bandwidth at the time of observation.

4. **Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.**
   Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.

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Source IP Address: 192.168.1.8

Destination IP Address: 172.217.194.110

Protocol: TLSv1.3

Packet Length: 571 bytes (observed packet)

Destination Port: 443

5. **Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.**

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AI-Generated Troubleshooting Checklist

1. Use an Ethernet Cable: Connect the gaming PC or console directly to the router using an Ethernet cable to reduce latency and packet loss.

2. Restart the Router: Restart the Wi-Fi router to refresh the network connection and resolve temporary connectivity issues.

3. Move Closer to the Router: If using Wi-Fi, sit closer to the router to get a stronger and more stable signal.

4. Close Background Applications: Close unnecessary downloads, streaming apps, and background programs that consume internet bandwidth.

5. Update Network Drivers: Keep the computer's Wi-Fi and Ethernet drivers updated to improve network performance.

6. Reduce Connected Devices: Disconnect unnecessary devices from the Wi-Fi network to reduce network congestion.

7. Check Internet Speed: Use an online speed test to check download speed, upload speed, and ping.

8. Update Router Firmware: Install the latest router firmware to improve performance, stability, and security.